

Frailty Policy Simulation: Method Details

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I. Verification & Validation (V&V)

1. Table 3 from the paper is reproduced below for a summary.

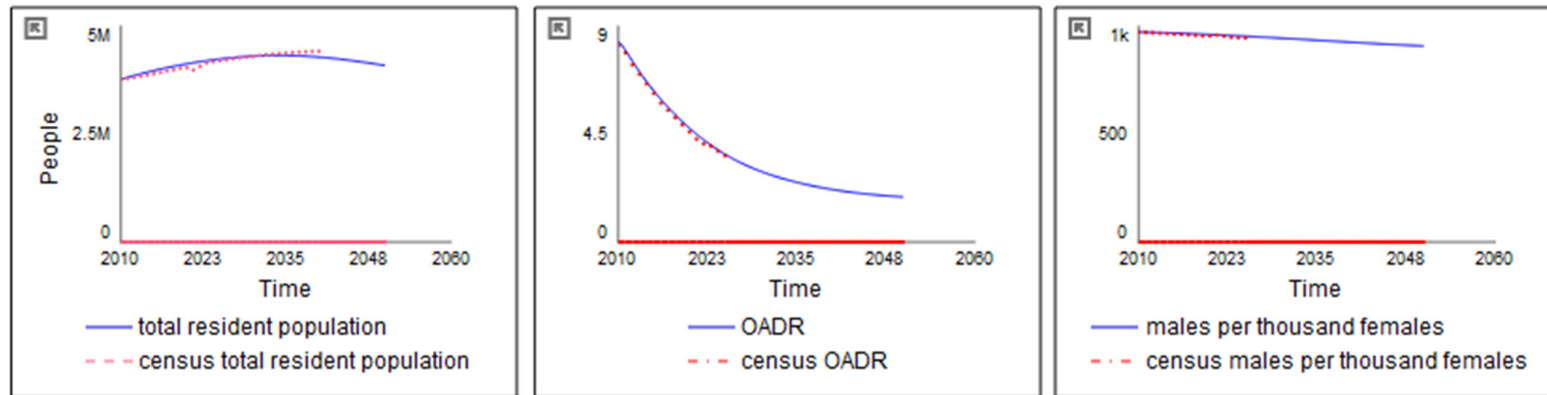
Table 3: V&V steps completed and planned.

	Meaning	Steps taken and planned
Conceptual validation	Model representation of the system is reasonable for the intended purpose	• Study team includes clinicians (BHR, LES) working in the frailty domain in Singapore. • Model logic, inputs, and results were presented to a group of practitioners dedicated to frailty management in June 2026. More reviews will be sought.
Operation validation	Output behavior has sufficient accuracy when compared to real-world	• Population outputs are validated against Singapore census data as well as available Department of Statistics projections for 2026-2040. • Frailty prevalence projections will be validated through further reviews with practitioners.
Specification verification	Software design and the specification for programming and implementing is satisfactory	• Model design has been reviewed by experts within the study team (RZ, SSWL) as well as an external expert.
Implementation verification	Simulation model is implemented according to the simulation model specification	• Model implementation has been reviewed within the team as well as an external expert. • The model is public and the study team welcomes review and improvement suggestions.

2. Population

- Fertility and mortality rates were based on 2010-2025 averages as per census data.
- Stella's model calibration feature was used to estimate immigration rate such that model total resident population aligned with census data for 2010-2025 as well as with the average of the lower and upper bound resident population projection for 2026-2040.

Screenshots from the model showing model versus official figures are shown below.



3. Frailty

- Frailty prevalence and incidence rates are based on Singapore studies.
- Some parameters (relative risk of mortality and of progression from pre-frail to frail for patients who are well-managed) were taken from global studies. Low and high values of these parameters (based on inter-quartile range) were used as factor settings in the analysis.
- Low and high values of 50% and 70% for the input variable for proportion of patients well-managed were based on discussion with a group of frailty practitioners.
- For further validation, the study team will continue to engage with stakeholders and seek current frailty prevalence statistics.